Sujata Sinha

Green Card Holder - No Sponsorship Required
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CAREER SUMMARY

Visiting Research Intern at Argonne National Laboratory and PhD Candidate at Virginia Tech with 5+ years of research experience across statistical machine learning, LLM pre-training and post-training, and scientific computing using statistical tools to drive data-driven decision making. Skilled at translating complex data into clear actionable recommendation that improves domain scientist, business and user experience.

EDUCATION

Ph.D. Candidate, Computer Engineering, Virginia Tech, VA. **December, 2026**

MS(Thesis), Computer Science, Auburn University, AL. **June, 2021**

B.Tech, Electronics Engineering, Uttar Pradesh Technical University, India. **June, 2018**

RESEARCH EXPERIENCE

Argonne National Laboratory & Virginia Tech Remote

Visiting Research Intern (Argonne) & Graduate Research Assistant (Virginia Tech) August 2024 – Present

Role: Collaborate with scientists, engineers, and DOE/NSF program managers and coordinate competing research priorities cross-functionally across academic and national laboratory partners to accelerate scientific discovery through AI..

- LLM Pre-Training, Post-Training and Scaling Laws
  - Established compute-optimal rules for LLM pre-training and post-training, validated against empirical scaling laws across 80 experiments with user-defined error bounds.
  - Derived information-theoretic framework linking model capacity to model retention, guiding data selection beyond entropy across a 20B-1T token corpus.
  - Built reproducible PyTorch pre- and post-training pipelines to support training for 70B+ models on 20B to 1T token corpus on exascale compute.
  - Cut training FLOPs by 50% at matched cross-entropy loss by optimizing parameter and token counts within the same model family and data distribution.
- LLM Post-Training and Modeling Alignment
  - Established optimal model-based data curation strategies validated across 10+ configurations.
  - Analyzed 1600+ LLM generated scientific hypothesis, identifying 82.1% non-viability rate and guiding post-training priorities that raised viability by 56% relative.
  - Designed and tested post-training strategies (SFT and RLHF) across 8 scientific tasks through 80+ controlled experiments, using expert-viability ratings and PCA to iselect the best alignment approach.
  - Post-trained a Llama4 judge for scientific hypothesis evaluation on 5B+ tokens per domain, spanning research articles, code, and scientific data.
- Large-Scale Multidimensional Data Analysis and Knowledge Compression
  - Established fundamental compression limits for multidimensional scientific data (images, videos, and 7D tensors), closing a 50+ year research gap through rate-distortion theory.
  - Built statistical modeling pipelines to characterize distributions and quantify information content across 33K+ multidimensional scientific data, guiding downstream training and compression decisions.
  - Validated applicability of fidelity-constrained bounds to 85% of the evaluated open-source scientific benchmark datasets by distributional checks, AIC/BIC model selection, and comparative experiments.

Virginia Tech Blacksburg, VA

Graduate Research Assistant August 2021 – July 2024

- Developed dual-discriminator GAN for data augmentation, reducing adversarial target by 43% across 6 attack conditions.
- Benchmarked 11-class neural networks and transformer models across 6 threat scenarios on a 220K+ evaluation subset, exposing attack failures under imperfect information.

Statistical Learning Lab, Auburn University Auburn, AL

Graduate Research Assistant August 2019 – July 2021

- Engineered TensorFlow pipeline to collect, analyze, and automate cardiac analysis across 100K+ MRI frames from 1000+ heterogeneous subjects, achieving 99.9% similarity scores against 90% baseline state-of-the-art.
- Designed subject-adaptive time-series features and benchmarked deep learning classifiers across 5 behavioral-state prediction tasks, outperforming the strongest conventional baseline by up to 7.45%.

TECHNICAL SKILLS

Programming: Python, PyTorch, TensorFlow, Numpy, SciPy, Scikit-learn, Pandas LaTeX
Research Methods: Hypothesis Testing, Uncertainty Estimation, Statistical Modeling, Machine Learning, Information Theory, Optimization
Data Visualization Tools: Matplotlib, Seaborn
Tools: Hugging Face, Exascale Computers (HPC)

LEADERSHIP EXPERIENCE

- Won Best Presentation Award at HPDC (acceptance rate <14%, awarded only 6 times in HPDC's 35-year history.
- Served as Virginia Tech's doctoral research lead on $3M DOE project.
- Reported quantitative results and findings on a weekly and quarterly basis to scientists and government stakeholders through technical reports and posters with actionable insights and recommendations to troubleshoot challenges.
- Presented AI and HPC research at selective conferences, communicating complex findings to multidisciplinary audience.
- Awarded Virginia Tech's Pratt Fellowship in recognition of excellence in graduate research.
- Awarded Auburn University's Graduate Fellowship Award in recognition of excellence in graduate research.

PUBLICATIONS

[1] **S. Sinha et. al.**, *Bridging Information Theory and Practice for Scientific Lossy Compression*, ACM HPDC (2026). **Best Presentation Award - awarded only 6 times in HPDC's 35-year history**.

[2] **S. Sinha et. al.**, *Rate-Distortion Bounds for Heterogeneous Random Fields on Finite Lattices*, (ArXiv:2603.09833) IEEE Trans. IT, Under Review.

[3] **S. Sinha et. al.**, *Which Post-Training Strategies Improve Scientific Hypothesis Viability?*, In Preparation.

[4] **S. Sinha et al.**, *Adversarial Attacks & Defenses for Wireless Signal Classifiers using CDI-aware GANs.* IEEE ICC, 2024.

[5] **S. Sinha et. al.**, *Channel Aware Adversarial Attacks are Not Robust.* IEEE MILCOM (2023).

[6] **Co-Author**, *Time-frequency analysis of scalp EEG with Hilbert-Huang transform and deep learning*, IEEE JBHI, 2021.

[7] **S. Sinha et. al.**, *Automated Semantic Segmentation of Cardiac Magnetic Resonance Images with Deep Learning.* IEEE ICMLA (2020).